

SINGH NITIN RAKESH

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Education

Uka Tarsadia University | Surat, Gujarat

B.Tech Artificial Intelligence & Data Science

Agg. CGPA: 8.68

Sep 2021 - May 2025

Achieved a perfect **10.0 SGPA** in the final semester, demonstrating mastery of advanced coursework

Technical Skills

Languages : Python(advanced), Javascript(basic), C/C++ (basic)

Tools & Technology : HTML, CSS, Node.js, Hugging Face, Power BI, Git, FastAPI, Flask, GitHub, Excel

Frameworks : Pandas, Numpy, Scikit-Learn, Matplotlib, Tensorflow, Keras, PyTorch

Technical Domains : Machine Learning, Deep Learning, OOP, Data Science, Computer Vision, NLP, GenAI, Manual Testing, IOT, MongoDB

Data Analysis & Visualization : SQL, Pandas, Numpy, Scikit-learn, Exploratory Data Analysis (EDA), Statistical Analysis, Data Cleaning

Databases & Cloud Platforms : Snowflake, Google Cloud Platform (GCP), AWS

Projects

Face Recognition Web Application (Find You) | RetinaFace, Facenet

[Github Link](#)

- This is a Flask-based web application for face recognition. Users can upload or capture photos to find matched faces from a dataset. The app also provides options to download all matched photos as a zip file.
- Ensuring Proper Security implementation by adding feature “Authorized access only by **Password-Protected-Link**”.
- **Achieved 96.3%** face recognition accuracy on a diverse dataset of 500+ images.

Solana Price Data Analysis | Data Engineering, Data Science, Data Analysis

[Github Link](#)

- **Analyzed** over **1,300 days** of historical market data to identify key drivers of price volatility and inform potential trading strategies.
- **Engineered** a robust data pipeline to clean, process, and enrich the raw dataset with **25+ technical indicators** (e.g., RSI, MACD), creating a comprehensive foundation for analysis
- **Synthesized** the enhanced 44-column dataset, which directly led to an **8% improvement** in the predictive performance of a baseline machine learning model.

Retail Sales Analysis on Snowflake | Snowflake, SQL

[Github Link](#)

- **Managed** an end-to-end data analysis pipeline within **Snowflake**, processing over **1 million rows** of Rossmann retail sales data to identify key drivers for demand forecasting.
- **Executed** data cleaning, preparation, and feature engineering **entirely with SQL**, handling missing values through imputation and creating new features like IsWeekend and PromoMonthFlag.
- **Uncovered** a critical business insight through Exploratory Data Analysis (EDA): promotional periods boosted average daily sales from EUR 4,406 to EUR 7,391, a remarkable **uplift of 67.7%**

RAG-Based Financial Intelligence Agent | Data Gen AI, RAG & Flask

[Github Link](#) | [Live Link](#)

- Developed a Retrieval-Augmented Generation (RAG) pipeline that aggregates real-time financial news from Polygon and Finviz APIs.
- Optimized the RAG pipeline using semantic chunking strategies and vector embeddings to manage context window limits, ensuring high-accuracy retrieval for Gemini Pro.
- Deployed the solution using Flask, handling dynamic context injection to ensure the LLM responses were grounded in live market data, eliminating hallucinations.

Multi-Agent Competitive Querying System (AMD-AI-League) | Agentic AI & LLM Orchestration

[Github Link](#)

- Architected a Multi-Agent System comprising distinct “Questioner” and “Answerer” agents to simulate competitive information retrieval scenarios.
- Implemented complex prompting strategies where the Question Agent generates context-aware queries and the Answer Agent retrieves and synthesizes data to respond.
- Designed the evaluation logic to score agent performance based on accuracy and relevance, laying the groundwork for automated negotiation systems.

Generative AI Image Pipeline & LoRA Fine-Tuning | Deep Learning & Computer Vision

[Github Link](#)

- Engineered an end-to-end image generation pipeline (AI-Avtaar) utilizing Stable Diffusion and Automatic1111 for high-fidelity character creation.
- Fine-tuned custom LoRA models and applied 4-bit Quantization techniques to optimize model inference latency and reduce memory usage during deployment.
- Integrated a Virtual Try-On module using Computer Vision (CatVTON), allowing users to overlay generated clothing onto custom avatars seamlessly.

Auto-Doc - VS Code Extension | JavaScript, VS Code API, Markdown, Git Diff Algorithm

[Github Link](#)

- Developed a zero-configuration VS Code extension that automatically tracks and documents code changes in real-time, creating time-stamped Markdown logs with precise line-by-line diffs (additions/removals with line numbers) to eliminate manual documentation effort for developers.
- Implemented intelligent diff detection algorithm that triggers on every file save, generating clean, readable change logs.

Experience

AI/ML Engineer ELaunch Solution Pvt. Ltd., Surat, Gujarat	Dec 2024 - Present
• Led the 'Solana Market Analysis' project, performing end-to-end exploratory data analysis (EDA) on over 1,300 days of historical data to identify key market drivers. • Engineered a data pipeline in Python to clean, process, and enrich the dataset with 25+ technical indicators (e.g., RSI, MACD), creating a comprehensive foundation for analysis. • Engineered and deployed an end-to-end AI image search system, performing rigorous model evaluation and tuning, achieving 92% accuracy.	
Data Analyst Intern JBcodeapp, Surat, Gujarat	May 2024 - June 2024
<i>Project: Student Performance Analysis Python, Numpy</i> • Engineered an automated data pipeline using Python to perform ETL (Extract, Transform, Load) on 5,000+ student records. • Conducted exploratory data analysis to identify the key factors and patterns influencing student academic performance. • Synthesized findings into a comprehensive report that isolated key factors influencing academic scores, providing a foundation for data-driven teaching methods and resource allocation strategies.	
AI/ML Trainee AlgoBrain AI, Surat, Gujarat	Sep 2023 - Jan 2024
<i>Project: Movie-Ticket Booking Chatbot with DialogFlow CX</i> • Analyzed potential user conversation paths to design and build a movie-ticket booking chatbot in DialogFlow CX, enhancing the overall user experience. • Systematically investigated 4 web applications to identify and resolve 30+ critical bugs, showcasing strong analytical skills in anomaly detection and root cause analysis.	

Extracurricular Activities

Research Community Member Swaayan Drone Club, SVNIT, Surat, Gujarat	May 2024 - June 2024
• Built and Assisted for Drone Bootcamps by building and calibrating drones using beta flight and mission planner.	
User Software Tester (Part-Time) Nothing India, Remote	Feb 2023 - Present
• Conducted user trial testing for Nothing Phone 1, Nothing Phone 2, Nothing 2a, Android 14, Cmf Watch, and more.	
Summer Analytics IIT Guwahati, Online	July 2024 - Aug 2024
• Successfully completed Summer Analytics 2024 Program with top 10 ranking out of all participants.	
Aspire Leaders Program Aspire Institute, Harvard Business School, Online	Aug 2024 - Nov 2024
• Completed all modules in Leadership, Communication, and Problem-Solving.	
AI Business Fellow Perplexity, Online	Feb 2025 - Aug 2025
• Selected in the Perplexity AI Business Fellow Program 2025.	
Gujarat State Lead Open Source Connect, Remote	Jul 2025 – Sep 2025
• Led Open Source Connect initiatives in Gujarat, driving AI, ML, and marketing outreach efforts.	
McKinsey.org Forward Program McKinsey.org, Online	Apr 2025 – Jun 2025
• Completed professional development in business analytics and strategic thinking	
Google Student Ambassador Program Google, Ping Network MCN, Online	Sep 2025 – Dec 2025
• Selected as the Google Student Ambassador from my college, serving as the campus representative for Google.	

Achievements

• Awarded a Certificate of Excellence in Summer Analytics (IIT Guwahati) for finishing in the top 10 of all participants. • Certified in Time Series Analysis by IIT Guwahati, with demonstrated expertise in statistical forecasting models. • Achieved Runner-Up position in Tattva Hackathon 2025 (CGPIT), demonstrating problem-solving, teamwork, and innovation. • Achieved Winner position in Quiz (Group) Competition at College Fest 2023, showcasing knowledge, teamwork, and quick thinking. • Secured 2nd Runner-Up position in Script Writing (Solo) at College Fest 2023, demonstrating creativity and communication skills.	
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Certifications

• Passed Google Analytics Certification. [View Credential] • Python for Data Science by NPTEL. [View Credential] • Completed AI/ML for GeoData Analysis by ISRO. [View Credential] • Completed ML: An Overview from Coursera. [View Credential]	
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Publications

Is Architectural Complexity Always the Answer? A Case Study on SwinIR vs. an Efficient CNN Chandresh Sutariya, Nitin Singh
• Abstract: Demonstrated that an optimized CNN achieves near state-of-the-art restoration quality comparable to SwinIR while significantly reducing computational overhead. • arXiv Pre-print, 2025